

Unifying Special Relativity and Quantum Mechanics

Lecture 13 — From Free to Interacting Theories

(Axioms, the Lagrangian Formalism, Noether's Theorem,
Canonical Quantization, and the Classification of Interactions)

Typed version of the handwritten notes, with added explanations in blue boxes

How to read this document

Conventions. Exactly as in Lectures 1–12: everything in the normal text and in the **yellow / pink** boxes is a faithful transcription of the handwritten notes (the coloured boxes reproduce the results that were framed or highlighted in the original — yellow boxes for the yellow-highlighted results, pink boxes for the pink/red-framed ones). Everything inside a **blue box like this one** is extra material that I added: derivations spelled out, terminology, corrections of transcription slips, and connections to earlier lectures.

Pagination of this scan. The notes open a *new chapter* and restart the page numbering from 1 for the **seventh** time. These sheets are therefore marked [p. 1*****]–[p. 22*****], with **seven stars**. There is no overlap with Lecture 12: the scan begins cleanly at a new section heading.

The lecture ends mid-topic. Page 22***** introduces the free Lagrangian for the massive spin-1 field and writes down the general quadratic ansatz, but stops before fixing the coefficients. That calculation presumably continues in the next lecture.

Where we left off. Lecture 12 introduced internal symmetries: a compact group G commuting with the Poincaré generators, realized à la Wigner–Weyl, and showed that demanding a charged field transform simply under $U(1)$ forces the existence of antiparticles. That lecture worked entirely with *free* fields built from creation and annihilation operators. This lecture asks what survives when interactions are switched on — and answers it by rebuilding the whole framework on the Lagrangian formalism.

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Part I — What survives when interactions are switched on

1 From free to interacting theories [p. 1*****]

What defines a general, interacting QFT?

Let us distinguish properties that are peculiar to free QFTs from properties that can be considered more universal.

■ Properties that are peculiar of a free theory

1) Decomposition in terms of creation and annihilation operators

$$\Phi^A(x) = \sum_{\lambda} \int \frac{d^3\vec{p}}{(2\pi)^3 \sqrt{2E_p}} \left[u^A(m, \vec{p}, \lambda) e^{-ip \cdot x} a(m, \vec{p}, \lambda) + v^A(m, \vec{p}, \lambda) e^{ip \cdot x} a^\dagger(m, \vec{p}, \lambda) \right]$$

Creation and annihilation operators act on the physical space of states by creating and removing individual particles with specific energy and momentum.

$\hookrightarrow$ Meaningful *only* in a free theory, since the concept of individual, non-interacting particles exists only in the absence of interactions.

2) Equations of motion like the KG equation and additional equations imposed on the free fields.

Example: for a massive spin-1 field, we find [p. 2*****]

$$(\square + M^2)V^\mu(x) = 0 ; \quad \partial_\mu V^\mu(x) = 0$$

$\hookrightarrow$ These are only valid in a free theory. In the context of an interacting theory, we expect these equations to change.

3) The Hamiltonian H_0 written in terms of creation and annihilation operators (similarly for $\vec{P}$, Q and other operators)

$$H_0 = \sum_m \sum_{\lambda} \int \frac{d^3\vec{p}}{(2\pi)^3} E_p a^\dagger(m, \vec{p}, \lambda) a(m, \vec{p}, \lambda)$$

Added explanation (not in the original notes)

Why this triage is the right way to start. Everything in Lectures 6–12 was built on the mode expansion: coefficient functions, polarization vectors, microcausality, the charge operator. None of it obviously survives interactions. So before writing a single interacting theory, one must ask: which statements were *about the physics* and which were artifacts of solvability? The three items above are the artifacts.

The deep reason item 1) fails. In an interacting theory the exact eigenstates of the full Hamiltonian are not free-particle states, and the field acting on the vacuum creates not a single particle but a superposition of one-particle, two-particle, ... states. Worse, *Haag's theorem* says the interacting field cannot be related to the free field by a unitary transformation at all — so there is no “interaction picture” in the naive sense. What remains meaningful is the *asymptotic* particle content at $t \rightarrow \pm\infty$, which is what scattering theory uses.

Why item 2) must change. $(\square + M^2)V^\mu = 0$ is linear; interactions by definition add terms with more fields, so the equation acquires a right-hand side. In fact the constraint $\partial_\mu V^\mu = 0$ is even more fragile: for the Proca field it followed from taking the four-divergence of the

equation of motion (Lecture 10, [p. 21****]), and once a source term is present that argument gives $\partial_\mu V^\mu \propto \partial_\mu J^\mu$ instead — which vanishes only if the current is conserved.

2 The state axioms and the field axioms [p. 2*****-4*****]

■ Properties that can be considered more general

► Properties of the states (“state axioms”)

1. Physical states transform as unitary representations of the Poincaré group (additionally supported by extra internal global symmetries).

↔ It implies that the theory must possess Hermitian operators $H, \vec{P}, J^{\mu\nu}, Q^A$ that generate the unitary representations.

2. There exists the vacuum, $|\Omega\rangle$, left invariant by the symmetries of the theory realized à la Wigner–Weyl:

$$H|\Omega\rangle = \vec{P}|\Omega\rangle = J^{\mu\nu}|\Omega\rangle = Q^A|\Omega\rangle = 0$$

3. The generators P^μ of space-time translations have a spectrum p^μ restricted to the forward light-cone:

$$p^0 \geq 0, \quad p^2 \geq 0$$

► Properties of the field operators (“field axioms”)

1. We have the following field transformation properties:

$$\begin{aligned}\Phi^A(x) &\longrightarrow U(\Lambda, a)^\dagger \Phi^A(x) U(\Lambda, a) = D(\Lambda)^A_B \Phi^B(\Lambda^{-1}x - \Lambda^{-1}a) \\ \Phi^a(x) &\longrightarrow U(\vec{\theta})^\dagger \Phi^a(x) U(\vec{\theta}) = D(\vec{\theta})^a_b \Phi^b(x)\end{aligned}$$

► $U(\Lambda, a)$ and $U(\vec{\theta})$ are unitary operators acting on the states.

At the infinitesimal level, the above transformations can be written in the form of commutators:

$$\begin{aligned}[P^\mu, \Phi^A(x)] &= -i\partial^\mu \Phi^A(x) \\ [J^{\mu\nu}, \Phi^A(x)] &= -\left[(J_S^{\mu\nu})^A_B + \vec{x} \times (-i\vec{\nabla})\delta^A_B\right]\Phi^B(x) \\ [Q^A, \Phi^a(x)] &= -(T^A)^a_b \Phi^b(x)\end{aligned}$$

↔ The generators $H, \vec{P}, J^{\mu\nu}, Q^A$ must verify the above commutators with the quantum fields.

2. Microcausality

$$\left[\Phi^A(x), \Psi^B(y)\right]_{\pm} = 0 \quad \text{if } (x - y)^2 < 0$$

Added explanation (not in the original notes)

Transcription note on the $J^{\mu\nu}$ commutator. The second line is written schematically: the orbital piece appears as $\vec{x} \times (-i\vec{\nabla})$, which is the familiar three-dimensional angular-momentum operator and so literally applies to the *rotation* components J^{ij} . The fully covariant statement, valid for all six components including the boosts, is

$$[J^{\mu\nu}, \Phi^A(x)] = - \left[(J_S^{\mu\nu})^A{}_B + i(x^\mu \partial^\nu - x^\nu \partial^\mu) \delta^A_B \right] \Phi^B(x),$$

whose spatial components reproduce the handwritten form. I have kept the original notation above and recorded the covariant version here.

What each axiom is doing. Notice how carefully the list avoids mentioning particles:

- *State axiom 1* keeps Wigner's classification — the reason mass and spin label particles at all.
- *State axiom 2* is the Wigner–Weyl condition of Lecture 12, now extended to the full symmetry group. The notation shifts from $|0\rangle$ to $|\Omega\rangle$ deliberately: in an interacting theory the true vacuum is *not* the free-theory Fock vacuum.
- *State axiom 3* is the spectral condition. $p^0 \geq 0$ says energies are non-negative; $p^2 \geq 0$ says no tachyons. Together they are what make the vacuum stable — without them one could lower the energy indefinitely.
- *Field axiom 1* says the fields are still tensors/spinors under Lorentz and still carry definite charges. Crucially, the commutator form survives interactions even though the mode expansion does not.
- *Field axiom 2* is microcausality — the same condition that fixed statistics in Lectures 7–10, here elevated to an axiom rather than derived from a mode expansion.

These are essentially the Wightman axioms. What is being sketched is the framework of axiomatic QFT: state a minimal list of physically motivated requirements, then ask what follows without ever solving a model. Several major results — the spin–statistics theorem, the CPT theorem, the Reeh–Schlieder theorem — are proved from exactly this list, with no perturbation theory anywhere.

Part II — The Lagrangian formalism and Noether's theorem

3 The Lagrangian formalism [p. 4*****-5*****]

The Lagrangian formalism, together with Noether's theorem, provides the right theoretical framework in which formulating interacting QFTs. We review the key points:

Consider a classical field theory described by the action functional

$$S = \int d^4x \mathcal{L}(\Phi^A(x), \partial_\mu \Phi^A(x))$$

► **Assumptions:** depends only on fields and derivatives (no higher order derivatives), all evaluated at the same space-time point (**locality**).

The Hamilton principle of stationary action gives the Euler–Lagrange E.o.M.:

$$\frac{\partial \mathcal{L}}{\partial \Phi^A} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \Phi^A)} = 0$$

one equation for each field component “A”

Consider now a continuous one-parameter field transformation

$$\Phi^A(x) \longrightarrow \Phi^A(x, \lambda), \quad \Phi^A(x, \lambda = 0) = \Phi^A(x)$$

Infinitesimally, we write

$$\Phi^A(x) \longrightarrow \Phi^A(x, \lambda) = \Phi^A(x, \lambda = 0) + \lambda \left. \frac{\partial \Phi^A(x, \lambda)}{\partial \lambda} \right|_{\lambda=0} = \Phi^A(x) + \lambda D\Phi^A(x)$$

where we identify:

$$\begin{aligned} \Phi^A(x) &\longrightarrow \Phi^A(x) + \lambda D\Phi^A(x) \\ D\Phi^A(x) &\equiv \left. \frac{\partial \Phi^A(x, \lambda)}{\partial \lambda} \right|_{\lambda=0} \end{aligned}$$

infinitesimal field transformation

Added explanation (not in the original notes)

Why the Lagrangian formalism is the right tool for interactions. Three reasons, all implicit in the page above:

- *It is constructive.* Instead of guessing equations of motion, one writes down the most general $\mathcal{L}$ compatible with the symmetries and *derives* them. Adding an interaction means adding a term to $\mathcal{L}$.
- *Symmetries are manifest.* A symmetry of $\mathcal{L}$ is visible by inspection, and Noether's theorem converts it automatically into a conserved current — no mode expansion required.

- *Locality is built in.* All fields at the same point x ; this is what will guarantee microcausality.

Why no higher derivatives. A Lagrangian containing $\partial^2\Phi$ or higher leads generically to equations of motion above second order, and hence (by Ostrogradsky's theorem) to a Hamiltonian unbounded from below — the energy can be made arbitrarily negative, and the vacuum is unstable. That directly contradicts state axiom 3. So the restriction is not for convenience: it is what keeps the theory physically sensible.

Reading $D\Phi^A$. The transformation is a whole curve in field space parametrized by λ , passing through the original field at $\lambda = 0$. $D\Phi^A$ is the tangent vector to that curve at the starting point — the “direction” in which the transformation moves the field. Everything that follows depends only on this first derivative, which is why infinitesimal transformations suffice.

4 Symmetries and Noether's theorem [p. 6*****-7*****]

We also have a change in field derivatives:

$$\partial_\mu \Phi^A(x) \longrightarrow \partial_\mu \Phi^A(x) + \lambda \partial_\mu D\Phi^A(x)$$

And we have a change in the Lagrangian density given by

$$\begin{aligned} \mathcal{L}(\Phi^A + \lambda D\Phi^A, \partial_\mu \Phi^A + \lambda \partial_\mu D\Phi^A) - \mathcal{L}(\Phi^A, \partial_\mu \Phi^A) &= \\ = \cancel{\mathcal{L}(\Phi^A, \partial_\mu \Phi^A)} + \frac{\partial \mathcal{L}}{\partial \Phi^A} \lambda D\Phi^A + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \Phi^A)} \lambda \partial_\mu D\Phi^A - \cancel{\mathcal{L}(\Phi^A, \partial_\mu \Phi^A)} \\ &\equiv \lambda D\mathcal{L} \end{aligned}$$

$$D\mathcal{L} = \frac{\partial \mathcal{L}}{\partial \Phi^A} D\Phi^A + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \Phi^A)} \partial_\mu (D\Phi^A)$$

infinitesimal change of the
Lagrangian density

The transformation $\Phi^A(x) \rightarrow \Phi^A(x, \lambda)$ is called a symmetry if

$$D\mathcal{L} = \partial_\mu F^\mu$$

the infinitesimal change of the Lagrangian
density is either zero or equals a
four-divergence of some scalar function F^μ

Obs.: λ is a parameter that does not depend on “ x ”. These are called global symmetries.

Noether theorem

For every infinitesimal global symmetry we have a conserved current $j^\mu(x)$ such that

$$\partial_\mu j^\mu(x) = 0$$

when the fields satisfy their E.o.M.

$$j^\mu(x) = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \Phi^A)} D\Phi^A - F^\mu$$

We also have conserved charges:

$$Q = \int d^3\vec{x} j^0(\vec{x}, t), \quad \frac{dQ}{dt} = 0$$

Added explanation (not in the original notes)

The proof, in three lines (not in the notes, but short enough to be worth having). Start from $D\mathcal{L}$ and use the Euler–Lagrange equation to replace $\partial\mathcal{L}/\partial\Phi^A = \partial_\mu[\partial\mathcal{L}/\partial(\partial_\mu\Phi^A)]$:

$$D\mathcal{L} = \partial_\mu \left[\frac{\partial \mathcal{L}}{\partial(\partial_\mu \Phi^A)} \right] D\Phi^A + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \Phi^A)} \partial_\mu (D\Phi^A) = \partial_\mu \left[\frac{\partial \mathcal{L}}{\partial(\partial_\mu \Phi^A)} D\Phi^A \right],$$

recognizing the product rule. If the transformation is a symmetry, $D\mathcal{L} = \partial_\mu F^\mu$, so $\partial_\mu \left[\frac{\partial \mathcal{L}}{\partial(\partial_\mu \Phi^A)} D\Phi^A - F^\mu \right] = 0$ — which is exactly the boxed current. Note where the E.o.M. entered: the current is conserved *on-shell* only.

Why $D\mathcal{L} = \partial_\mu F^\mu$ and not just $D\mathcal{L} = 0$. The action, not the Lagrangian, is what physics depends on. A four-divergence integrates to a boundary term, $\int d^4x \partial_\mu F^\mu = \oint dS_\mu F^\mu$, which vanishes for fields decaying at infinity — so S is unchanged and the equations of motion are unchanged. Allowing F^μ is not a technicality: space-time translations are exactly of this type, since $\mathcal{L}$ itself shifts by a total derivative under $x \rightarrow x + a$.

From current to charge. Integrate $\partial_\mu j^\mu = 0$ over all space:

$$\frac{dQ}{dt} = \int d^3\vec{x} \partial_0 j^0 = - \int d^3\vec{x} \vec{\nabla} \cdot \vec{j} = - \oint_{\text{surface at } \infty} d\vec{S} \cdot \vec{j} = 0,$$

assuming the current falls off fast enough. Conservation of Q is thus the statement that nothing leaks out through the boundary — the local statement $\partial_\mu j^\mu = 0$ is strictly stronger than the global one.

Why “global”. If λ were allowed to depend on x , the derivative $\partial_\mu(\lambda D\Phi)$ would produce an extra term $(\partial_\mu \lambda) D\Phi$ that spoils the argument. Making a global symmetry local is exactly the gauging procedure — which requires introducing a gauge field, as the massless constructions of Lecture 11 already hinted.

5 Example: a compact Lie group G [p. 7*****–9*****]

Consider a set of fields that transform according to some compact Lie group G . We write:

$$\begin{aligned} \Phi^a(x) &\longrightarrow \Phi'^a(x) = \left(\exp(-i\theta_A T^A) \right)_b^a \Phi^b(x) \\ &= \left[\delta_b^a - i\theta_A (T^A)_b^a + O(\theta^2) \right] \Phi^b(x) \\ &= \Phi^a(x) - i\theta_A (T^A)_b^a \Phi^b(x) \end{aligned}$$

θ_A is a set of $\dim G$ independent parameters

We extract

$$(D\Phi^a)^A = -i(T^A)^a_b \Phi^b(x) \quad \text{for } A = 1, \dots, \dim G$$

infinitesimal change of the field in the “direction” A

Suppose that we now construct the Lagrangian in such a way that it remains invariant under the above transformation, $D\mathcal{L} = 0$.

↔ We thus have a symmetry and, as a consequence, the presence of $\dim G$ conserved currents and charges:

$$j^{\mu A} = \frac{\partial \mathcal{L}}{\partial (\partial_\mu \Phi^a)} (-i)(T^A)^a_b \Phi^b(x)$$

$$Q^A \equiv \int d^3\vec{x} j^{0A}(\vec{x}, t)$$

We learn an important lesson: for a given field content, if we impose that a certain transformation is a symmetry, we get constraints that a Lagrangian density must satisfy.

↔ In this case, the Lagrangian density must be invariant, $D\mathcal{L} = 0$.

This is interesting since it implies that we can shape the form of a Lagrangian density by imposing symmetries.

Added explanation (not in the original notes)

This is the design principle of the Standard Model. One does not guess the Lagrangian of particle physics; one specifies the field content and the symmetry group, then writes the most general $\mathcal{L}$ consistent with both (up to a chosen mass dimension — the classification on [p. 20*****]–[p. 21*****] below). For the Standard Model the input is $SU(3)_c \times SU(2)_L \times U(1)_Y$ plus a list of representations, and essentially everything else — the interactions, their relative strengths, the conservation laws — follows.

Why $F^\mu = 0$ here. For internal symmetries the fields at x are rotated among themselves but the point x is untouched, so a Lagrangian built from invariant combinations (e.g. $\Phi^\dagger \Phi$) changes by exactly nothing rather than by a total derivative. Contrast space-time symmetries on [p. 15*****], where $F^\mu \neq 0$ and the current is the energy–momentum tensor.

Consistency with Lecture 12. There, $[Q^A, \Phi^a] = -(T^A)^a_b \Phi^b$ was obtained by comparing the unitary action on states with the representation matrix on fields. Here Q^A is instead *constructed* from the Lagrangian via Noether’s theorem. Pages 11*****–12***** verify that the two definitions agree — which is the real content of the “closing of the circle” on [p. 14*****].

Part III — Canonical quantization and the Noether charges

6 Canonical quantization [p. 9*****-10*****]

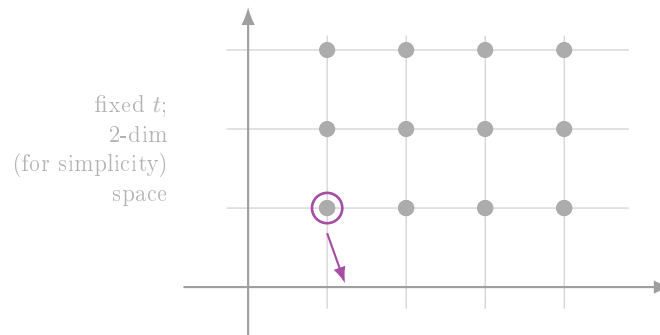
The discussion so far has been classical. We now quantize the theory. We introduce conjugate momenta

$$\Pi_A \equiv \frac{\partial \mathcal{L}}{\partial(\partial_0 \Phi^A)}$$

and impose equal-time commutation / anticommutation relations:

$$\begin{aligned} \left[\Phi^A(t, \vec{x}), \Phi^B(t, \vec{y}) \right]_{\pm} &= 0 \\ \left[\Pi_A(t, \vec{x}), \Pi_B(t, \vec{y}) \right]_{\pm} &= 0 \\ \left[\Phi^A(t, \vec{x}), \Pi_B(t, \vec{y}) \right]_{\pm} &= i \delta(\vec{x} - \vec{y}) \delta^A_B \end{aligned}$$

We can think about these relations as the continuous limit of quantum particle mechanics.



At each point in the lattice:

$$\begin{cases} \phi(t, \vec{x}_i) \equiv q_{\vec{x}_i}(t) \\ \Pi(t, \vec{x}_i) \equiv p_{\vec{x}_i}(t) \end{cases} \quad \text{with} \quad \left[q_{\vec{x}}(t), p_{\vec{y}}(t) \right] = i \delta_{\vec{x}\vec{y}}$$

↔ In the continuous limit:

$$\left[\phi(t, \vec{x}), \Pi(t, \vec{y}) \right] = i \delta(\vec{x} - \vec{y})$$

with the additional requirement that, in the case of fields describing spin half-integer, we have anticommutators.

After (canonical) quantization, we highlight the following properties:

Added explanation (not in the original notes)

Transcription note. The margin annotation next to the figure is clipped at the left edge of the scan; it reads “fixed t ; 2-dim (for simplicity) space” — i.e. the picture is a snapshot at one instant, drawn in two spatial dimensions instead of three purely for drawing convenience.

The lattice picture, unpacked. Replace continuous space by a grid of points spaced a apart. At each site sits one degree of freedom: a value of the field $q_{\vec{x}}$ and its conjugate momentum $p_{\vec{x}}$. This is just ordinary quantum mechanics of a large number of coupled oscillators — nothing new. As $a \rightarrow 0$ the Kronecker delta becomes a Dirac delta, $\delta_{\vec{x}\vec{y}}/a^3 \rightarrow \delta^{(3)}(\vec{x} - \vec{y})$, and one recovers the field commutator. So canonical quantization of a field is not a new postulate: it is the familiar $[q, p] = i$ applied to infinitely many degrees of freedom, one per point of space.

Why equal time. The commutators are imposed at a single instant because that is where the canonical structure lives — Φ and Π are the coordinates and momenta on a fixed time slice. The equations of motion then propagate them to all other times. Note that this is manifestly *not* Lorentz covariant in appearance (a time slice is frame-dependent), yet the resulting theory is covariant. Microcausality is the statement that reconciles the two.

The half-integer caveat. As always in this course, anticommutators are not an extra assumption but the outcome of the spin–statistics argument (Lectures 7–10): the spin sum for half-integer spin is odd under $p \rightarrow -p$, which flips the relative sign and makes only the anticommutator vanish outside the light cone.

7 The Noether charges as operators [p. 11 *****-12*****]

1) We promote the Noether charges as follows:

$$Q^A = \int d^3\vec{x} \frac{\partial \mathcal{L}}{\partial(\partial_0 \Phi^a)} (-i)(T^A)^a_b \Phi^b(x) = (-i) \int d^3\vec{x} \Pi_a(x) (T^A)^a_b \Phi^b(x)$$

$$Q^A = \int d^3\vec{x} (-i) \Pi_a(x) (T^A)^a_b \Phi^b(x)$$

$\hookrightarrow$ promoted to be operators

2) We compute $[Q^A, \Phi^a(x)]$:

$$[Q^A, \Phi^a(t, \vec{x})] = \int d^3\vec{y} \left[(-i) \Pi_c(t, \vec{y}) (T^A)^c_b \Phi^b(t, \vec{y}), \Phi^a(t, \vec{x}) \right]$$

$\hookrightarrow$ since Q^A is conserved, we evaluate it at the same time “ t ” of the field

$$= \int d^3\vec{y} \left[\underbrace{\Pi_c(t, \vec{y})}_{\equiv A} \underbrace{(-i)(T^A)^c_b \Phi^b(t, \vec{y})}_{\equiv B}, \underbrace{\Phi^a(t, \vec{x})}_{\equiv C} \right]$$

We use $[AB, C] = A[B, C]_{\pm} \mp [A, C]_{\pm} B$:

$$= \int d^3\vec{y} (\mp) \left[\Pi_c(t, \vec{y}), \Phi^a(t, \vec{x}) \right]_{\pm} (-i)(T^A)^c_b \Phi^b(t, \vec{y})$$

Since $[\Pi_c(t, \vec{y}), \Phi^a(t, \vec{x})]_{\pm} = \pm [\Phi^a(t, \vec{x}), \Pi_c(t, \vec{y})]_{\pm}$,

$$= \int d^3\vec{y} (-1) i \delta^a_c \delta(\vec{x} - \vec{y}) (-i)(T^A)^c_b \Phi^b(\vec{y}, t) = -(T^A)^a_b \Phi^b(t, \vec{x})$$

We find

$$[Q^A, \Phi^a(x)] = -(T^A)^a_b \Phi^b(x)$$

A completely analogous computation gives $[Q^A, \Pi_a(t, \vec{x})]$:

$$\begin{aligned} [Q^A, \Pi_a(t, \vec{x})] &= \int d^3\vec{y} [\Pi_c(t, \vec{y})(-i)(T^A)^c_b \Phi^b(t, \vec{y}), \Pi_a(t, \vec{x})] \\ &= \int d^3\vec{y} \Pi_c(t, \vec{y})(-i)(T^A)^c_b [\Phi^b(t, \vec{y}), \Pi_a(t, \vec{x})]_{\pm} \\ &= \int d^3\vec{y} \Pi_c(t, \vec{y})(-i)(T^A)^c_b i \delta^b_a \delta(\vec{x} - \vec{y}) \\ &= +\Pi_c(t, \vec{x})(T^A)^c_a \end{aligned}$$

$$[Q^A, \Pi_a(x)] = \Pi_b(x)(T^A)^b_a$$

Added explanation (not in the original notes)

Why the charge may be evaluated at the field's time. Q^A is time-independent, so we are free to choose the instant at which the spatial integral is performed. Choosing t equal to the time argument of the field is what lets the *equal-time* canonical relations be used — otherwise the commutator would require knowing the full time evolution. This small step is what makes the whole computation possible.

Statistics-independence, again. As in Lecture 12, the identity $[AB, C] = A[B, C]_{\pm} \mp [A, C]_{\pm}B$ holds for both signs, and here the term $[\Phi, \Phi]_{\pm} = 0$ vanishes either way while the surviving canonical bracket has the same value. The result is therefore identical for bosons and fermions — worth noting, since T^A acts on internal indices and should not care about spin.

The result reproduces the field axiom. Compare $[Q^A, \Phi^a] = -(T^A)^a_b \Phi^b$ with the third commutator listed on [p. 4*****] as an *axiom*, and with the same relation derived in Lecture 12 from the mode expansion. Three independent routes, one answer. That agreement is the point of the whole section.

The sign pattern for Π . Φ carries an upper index and transforms with $-T^A$; Π carries a lower index and transforms with $+T^A$ acting from the right. This is exactly how a representation and its *dual* (conjugate) representation transform — as it must be, since $\Pi_a \Phi^a$ is a scalar and its commutator with Q^A must vanish.

8 The charge algebra closes [p. 13*****–14*****]

3) We now compute $[Q^A, Q^B]$:

$$\begin{aligned} [Q^A, Q^B] &= \int d^3\vec{x} [Q^A, \Pi_a(t, \vec{x})(-i)(T^B)^a_b \Phi^b(t, \vec{x})] \\ &= \int d^3\vec{x} \left\{ [Q^A, \Pi_a(x)](-i)(T^B)^a_b \Phi^b(x) + \Pi_a(x)(-i)(T^B)^a_b [Q^A, \Phi^b(x)] \right\} \\ &= \int d^3\vec{x} \left\{ \Pi_b(x)(T^A)^b_a (-i)(T^B)^a_c \Phi^c(x) + \Pi_a(x)(-i)(T^B)^a_b (-1)(T^A)^b_c \Phi^c(x) \right\} \end{aligned}$$

$$\begin{aligned}
&= (-i) \int d^3\vec{x} \left\{ \Pi_b(x) (T^A T^B)^b_a \Phi^a(x) - \Pi_b(x) (T^B T^A)^b_a \Phi^a(x) \right\} \\
&= (-i) \int d^3\vec{x} \Pi_b(x) \underbrace{\left([T^A, T^B] \right)^b_a}_{[T^A, T^B] = if_{ABC} T^C} \Phi^a(x) \\
&= if_{ABC} (-i) \underbrace{\int d^3\vec{x} \Pi_b(x) (T^C)^b_a \Phi^a(x)}_{\text{this is } Q^C} = if_{ABC} Q^C
\end{aligned}$$

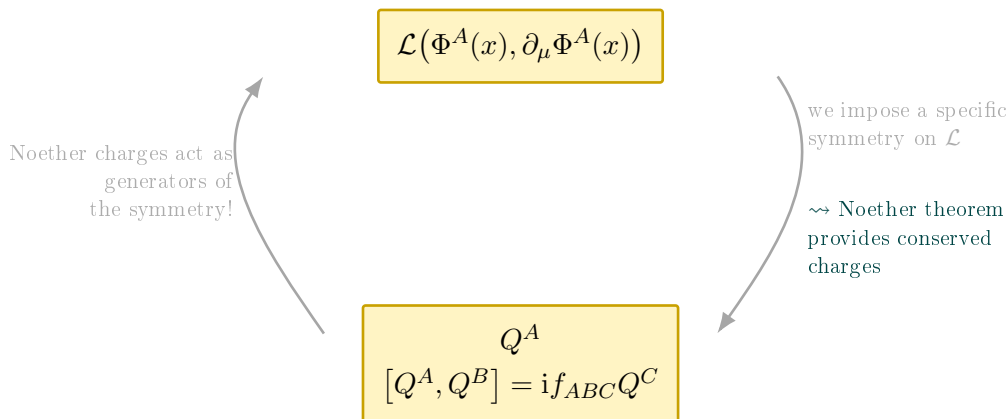
That is

$$[Q^A, Q^B] = if_{ABC} Q^C$$

$\hookrightarrow$ The Noether charges close the algebra of the group G .

Let us now comment about our results: the following “closing of the circle” emerges.

Lagrangian for a given field content



We thus use Noether’s theorem as follows:

Added explanation (not in the original notes)

Why the closing of the circle is not a triviality. The two arrows are logically independent operations. Going up: pick a symmetry, apply Noether’s theorem, obtain some integrals of field operators. Going down: ask whether those integrals *generate* the symmetry they came from — i.e. whether $e^{i\theta_A Q^A}$ implements the very transformation used to derive them, and whether the Q^A reproduce the Lie algebra of G . Nothing guarantees this in advance; it is the content of the calculations on the previous pages.

What it buys us. The state axioms of [p. 2*****] demanded the existence of Hermitian operators $H, \vec{P}, J^{\mu\nu}, Q^A$ generating unitary representations. We now have a *recipe* for producing them from any Lagrangian — with no mode expansion, hence valid in the interacting theory. That is precisely the bridge the lecture set out to build.

Why if_{ABC} and not something else. The structure constants that appear are those of G itself, not of the particular representation T^A : different fields in different representations give the *same* algebra for the charges. This is what allows charges from different sectors of a theory to be added together — total electric charge, total isospin — and is the operator counterpart of the abstract group structure.

9 Space-time symmetries: the Poincaré charges [p. 15*****-17*****]

The Lagrangian is invariant under a global symmetry group G

$\Rightarrow$

Noether charges Q^A that generate the symmetry in the sense we discussed

$$\mathcal{L}(\Phi^A(x), \partial_\mu \Phi^A(x))$$

The Lagrangian does not depend explicitly on “ x ”

$\Downarrow$

space-time translations is a symmetry

$\Downarrow$

the Noether charges are $H \equiv P^0, \vec{P}$

The Lagrangian is a scalar under Lorentz

$\Downarrow$

Lorentz transformations is a symmetry

$\Downarrow$

the Noether charges are $J^{\mu\nu}$

- The charges $H, \vec{P}, J^{\mu\nu}$ close the algebra of the Poincaré group.
- The charges $H, \vec{P}, J^{\mu\nu}$ generate the correct symmetry transformations of the fields.
- If the Lagrangian is Hermitian, the charges are Hermitian too (this is important since we want them to generate a unitary representation).

Example (review of known things)

In the case of space-time translations, the conserved currents are

$$T^{\mu\nu} = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \Phi^A)} (\partial^\nu \Phi^A) - g^{\mu\nu} \mathcal{L}$$

canonical energy–momentum tensor

$\hookrightarrow$ The four Noether charges are

$$P^\mu \equiv \int d^3 \vec{x} T^{0\mu}$$

The $\mu = 0$ charge is

$$P^0 = \int d^3 \vec{x} \left[\frac{\partial \mathcal{L}}{\partial(\partial_0 \Phi^A)} (\partial^0 \Phi^A) - g^{00} \mathcal{L} \right] = \int d^3 \vec{x} \left[\Pi_A (\partial^0 \Phi^A) - \mathcal{L} \right] = \int d^3 \vec{x} \left(\Pi_A \dot{\Phi}^A - \mathcal{L} \right)$$

this is the Legendre transform that gives the Hamiltonian density

$$P^0 \equiv H = \int d^3 \vec{x} \underbrace{\left[\Pi_A(t, \vec{x}) \dot{\Phi}^A(t, \vec{x}) - \mathcal{L}(\Phi^A, \partial_\mu \Phi^A) \right]}_{\equiv \mathcal{H}(x) \text{ Hamiltonian density}}$$

this is the “full” Hamiltonian

After quantization, we also find

$$\begin{aligned}[H, \Phi^A(x)] &= -i\dot{\Phi}^A(x) \\ [H, \Pi_A(x)] &= -i\dot{\Pi}_A(x)\end{aligned}$$

which imply that H generates time-translation transformations.

Added explanation (not in the original notes)

Why “full” Hamiltonian matters. Contrast with H_0 on [p. 2*****], which was built from creation and annihilation operators and counted free particles. The H here is constructed from the Lagrangian, so it automatically includes every interaction term. It is the operator that actually governs time evolution in the interacting theory — and it is the one that must satisfy the state axioms.

A caveat on $T^{\mu\nu}$ worth knowing. The *canonical* energy-momentum tensor above is not in general symmetric in $\mu \leftrightarrow \nu$, which is awkward because the angular-momentum current requires a symmetric tensor and because coupling to gravity requires one. The standard fix is the Belinfante–Rosenfeld improvement: add a term $\partial_\rho B^{\rho\mu\nu}$ with $B^{\rho\mu\nu} = -B^{\mu\rho\nu}$, which changes neither the conservation law nor the charges P^μ but symmetrizes the tensor. For a scalar field the canonical tensor is already symmetric; for the vector and spinor fields of this course it is not.

Reading $[H, \Phi] = -i\dot{\Phi}$. This is the Heisenberg equation of motion $\dot{\mathcal{O}} = i[H, \mathcal{O}]$, and it is the $\mu = 0$ component of the first field axiom on [p. 4*****], $[P^\mu, \Phi^A] = -i\partial^\mu \Phi^A$. Once again a Lagrangian-derived charge is seen to satisfy an axiom that was previously simply postulated.

10 Finite time translations and the Heisenberg picture [p. 18*****]

Obs.: The above relations correspond to infinitesimal time translations. It is however possible to write finite time-translations as follows:

$$e^{iHt} \Phi(t=0, \vec{x}) e^{-iHt} = ?$$

Lemma di Hadamard:

$$e^A B e^{-A} = B + [A, B] + \frac{1}{2!} [A, [A, B]] + \frac{1}{3!} [A, [A, [A, B]]] + \dots$$

$$\begin{aligned}e^{iHt} \Phi(t=0, \vec{x}) e^{-iHt} &= \Phi(t=0, \vec{x}) + it[H, \Phi(t=0, \vec{x})] + \frac{(it)^2}{2!} [H, [H, \Phi(t=0, \vec{x})]] + \dots \\ &= \Phi(t=0, \vec{x}) + it(-i)\dot{\Phi}(t=0, \vec{x}) + \frac{(it)^2}{2!} (-i) \underbrace{[H, \dot{\Phi}(t=0, \vec{x})]}_{(-i)\ddot{\Phi}(t=0, \vec{x})} + \dots \\ &= \Phi(t=0, \vec{x}) + t\dot{\Phi}(t=0, \vec{x}) + \frac{t^2}{2!} \ddot{\Phi}(t=0, \vec{x}) + \dots \\ &= \Phi(t, \vec{x})\end{aligned}$$

$$\Phi(t, \vec{x}) = e^{iHt} \Phi(t=0, \vec{x}) e^{-iHt}$$

The operator $\Phi(t, \vec{x})$ evolves in time according to the **full** Hamiltonian.

- In a fully interacting QFT, field operators are in the Heisenberg picture.

Added explanation (not in the original notes)

What the calculation really is. The last line is nothing but the Taylor expansion of $\Phi(t, \vec{x})$ about $t = 0$. So the statement “conjugating by e^{iHt} translates in time” is equivalent to “ H generates the Taylor series of the time evolution” — the finite version of $[H, \Phi] = -i\dot{\Phi}$. The Hadamard lemma is what resums the infinitesimal statement into a finite one.

Why the Heisenberg picture is forced. In elementary quantum mechanics one may choose Schrödinger, Heisenberg, or interaction picture freely. In an interacting QFT the choice is not free: the interaction picture requires splitting $H = H_0 + H_{\text{int}}$ and relating the interacting field to a free one by a unitary transformation — exactly what Haag’s theorem forbids. The Heisenberg picture, where the operators carry the full time dependence and the states are fixed, is the one that survives.

Consistency with the field axioms. The relation $\Phi(x) = e^{iHt}\Phi(0, \vec{x})e^{-iHt}$ is the $\mu = 0$ case of the general covariant statement $\Phi^A(x) = e^{iP \cdot x}\Phi^A(0)e^{-iP \cdot x}$, which is how the first field axiom looks in finite form. All the x -dependence of a quantum field is generated by the translation operators.

Part IV — Building Lagrangians: dimensions and interactions

11 Further properties of a Lagrangian density [p. 19*****]

There are additional criteria to build Lagrangian densities.

■ Mass dimension

In natural units, the Lagrangian has mass dimension 4:

$$[\mathcal{L}] = [M]^4$$

■ Expansion in fields and derivatives

We organize Lagrangian densities as an expansion in fields and derivatives.

1) The first order in this expansion contains terms quadratic in the fields and terms that are at most quadratic in derivatives of the fields.

↔ This is the free theory.

The free theory must reproduce all relevant physics that we previously discussed.

2) Terms in which we have more than two fields are known as interaction terms.

Added explanation (not in the original notes)

Where $[\mathcal{L}] = [M]^4$ comes from. The action $S = \int d^4x \mathcal{L}$ must be dimensionless (it appears in $e^{iS/\hbar}$ and $\hbar = 1$). Since $[d^4x] = [M]^{-4}$, we need $[\mathcal{L}] = [M]^4$. Everything in the classification that follows is a consequence of this one constraint.

The dimensions of the fields, collected from earlier lectures: a scalar has $[\phi] = [M]$, a vector $[V^\mu] = [M]$ (Lecture 10, [p. 24****]), a spinor $[\Psi] = [M]^{3/2}$, and each derivative $[\partial_\mu] = [M]$. One can check these directly from the kinetic terms: e.g. $(\partial\phi)^2$ has dimension $2 + 2[\phi] = 4$, giving $[\phi] = 1$, while $\bar{\Psi}\gamma^\mu\partial_\mu\Psi$ gives $2[\Psi] + 1 = 4$, i.e. $[\Psi] = 3/2$.

Why the free theory is exactly the quadratic part. Quadratic terms give *linear* equations of motion, which is the definition of a free theory: solutions superpose, and the theory can be solved exactly by Fourier transform. The moment a cubic or higher term appears, the equations become non-linear and modes start talking to each other — interactions.

12 The classification of interactions [p. 20*****–22*****]

We classify interactions into three categories:

1) Relevant interactions (a.k.a. “super-renormalizable”)

Interactions consisting of operators that, based solely on the powers of fields and derivatives, have a **mass dimension less than 4**.

Example

$$\mathcal{L}_{\text{int}} = \mu \Phi(x)^3$$

► $[\Phi] = [M]$ (real scalar field). If we only count the number of fields, the dimension is $3 < 4$. Since $[\mathcal{L}] \stackrel{!}{=} [M]^4$, the coefficient “ μ ” must have a positive mass dimension, in this case $[\mu] = [M]$.

2) Marginal interactions (a.k.a. “renormalizable”)

Interactions consisting of operators that, based solely on the powers of fields and derivatives, have a **mass dimension equal to 4**.

Example

$$\mathcal{L}_{\text{int}} = g V_\mu(x) \bar{\Psi}(x) \gamma^\mu \Psi(x)$$

► $[\Psi] = [M]^{3/2}$; $[V^\mu] = [M]$. If we only count the number of fields, the dimension is 4. Since $[\mathcal{L}] \stackrel{!}{=} [M]^4$, the coefficient in front is dimensionless.

3) Irrelevant interactions (a.k.a. “non-renormalizable”)

Interactions consisting of operators that, based solely on powers of fields and derivatives, have a **mass dimension larger than 4**.

Example

$$\mathcal{L}_{\text{int}} = \kappa F_{\mu\nu}(x) \bar{\Psi}(x) \sigma^{\mu\nu} \Psi(x)$$

► $[\Psi] = [M]^{3/2}$; $[F_{\mu\nu}] = [M]^2$. If we count the number of fields and derivatives, the dimension is $5 > 4$. Consequently, since $[\mathcal{L}] \stackrel{!}{=} [M]^4$, the coefficient in front must have negative mass dimension, in this case $[\kappa] = [M]^{-1}$.

Added explanation (not in the original notes)

Why the names “relevant”, “marginal”, “irrelevant”. They come from the renormalization group and refer to behaviour at *low* energies. Write the coupling of an operator of dimension d as c/Λ^{d-4} with Λ some high scale. At an energy E the dimensionless effective strength is $\sim (E/\Lambda)^{d-4}$:

- $d < 4$: grows as E decreases — *relevant* at low energy, and increasingly important in the infrared.
- $d = 4$: energy-independent at leading order — *marginal*; the running is only logarithmic and is decided by quantum corrections.
- $d > 4$: shrinks as E decreases — *irrelevant* at low energy, suppressed by powers of E/Λ .

Why the other names, “renormalizable” etc. A theory containing only $d \leq 4$ operators requires finitely many counterterms and is predictive to all orders. Once $d > 4$ operators are present, each loop order demands new counterterms and infinitely many parameters are needed — traditionally regarded as fatal.

The modern reading: effective field theory. Non-renormalizable is no longer a death sentence. An irrelevant operator with $[\kappa] = [M]^{-1}$ simply signals a new scale $\Lambda \sim 1/\kappa$ above which the description breaks down; below it the theory is perfectly predictive to any desired accuracy in E/Λ . Two famous instances: Fermi’s four-fermion theory of the weak

interaction has $[G_F] = [M]^{-2}$ and pointed to the W boson; and general relativity itself is non-renormalizable, with $[G_N] = [M]^{-2}$ pointing to the Planck scale. The example above, the magnetic-moment operator $\kappa F_{\mu\nu} \bar{\Psi} \sigma^{\mu\nu} \Psi$, is precisely how anomalous magnetic moments are parametrized in searches for new physics.

The phrase “based solely on the powers of fields and derivatives” is doing real work. It instructs us to count the dimension of the *operator*, ignoring the coupling in front — because the coupling’s dimension is then *determined* by the requirement $[\mathcal{L}] = [M]^4$. The sign of that determined dimension is what sorts the operator into one of the three classes.

13 The massive spin-1 field: the free theory [p. 22*****]

Consider the field $V^\mu(x)$. As a free field, it satisfies the equations

$$(\square + M^2)V^\mu(x) = 0, \quad \partial_\mu V^\mu(x) = 0.$$

We look for a Lagrangian description.

↔ For the free theory, we look for a Lagrangian with terms quadratic in the fields and at most quadratic in derivatives. We take into account that $[V^\mu] = [M]$.

We write

$$\mathcal{L} = -\frac{\alpha}{2}(\partial_\mu V_\nu)(\partial^\mu V^\nu) - \frac{\beta}{2}(\partial_\mu V_\nu)(\partial^\nu V^\mu) + \frac{1}{2}M^2 V_\mu V^\mu$$

* * *

(End of the Lecture 13 scan — the coefficients α, β are fixed on the next sheet.)

Added explanation (not in the original notes)

Why this is the most general ansatz. We need terms quadratic in V^μ with at most two derivatives, Lorentz invariant, and of total mass dimension 4. With $[V] = [M]$, a term with two fields and two derivatives already has dimension 4, so its coefficient is dimensionless; a term with two fields and no derivatives has dimension 2, so its coefficient carries $[M]^2$ — which is the mass term. The only ways to contract two derivatives and two vector indices are

$$(\partial_\mu V_\nu)(\partial^\mu V^\nu), \quad (\partial_\mu V_\nu)(\partial^\nu V^\mu), \quad (\partial_\mu V^\mu)(\partial_\nu V^\nu),$$

and the third differs from the second by a total derivative, so it need not be listed separately. Hence two independent structures α, β plus the mass term — exactly the ansatz written.

Where it is going (a preview). Demanding that the Euler–Lagrange equations reproduce *both* $(\square + M^2)V^\mu = 0$ and the constraint $\partial_\mu V^\mu = 0$ fixes the ratio. The answer is $\beta = -\alpha$, with the overall normalization $\alpha = 1$ chosen conventionally, which assembles the two derivative terms into

$$-\frac{1}{2}(\partial_\mu V_\nu)(\partial^\mu V^\nu - \partial^\nu V^\mu) = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu}, \quad F_{\mu\nu} = \partial_\mu V_\nu - \partial_\nu V_\mu,$$

giving the Proca Lagrangian $\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}M^2 V_\mu V^\mu$ anticipated in Lecture 10 ([p. 21****]). The point of the exercise is that this famous Lagrangian is not guessed: it is the *unique* outcome of the requirements “quadratic, at most two derivatives, Lorentz invariant, correct free equations”.

End of Lecture 13.